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Week 11 assignment

Databricks has a strong commitment to open source community. The following technologies are open source projects founded by Databricks employees:

Delta Lake

Delta Sharing

ML flow

Apache Spark

Redash

Databricks maintains a number of proprietary tools that integrate and expand these technologies to add optimized performance and ease of use, such as the following:

Workflow

Unity Catalog

Delta Live tables

Databricks SQL

Photon

You configure a Databricks workspace by configuring secure integrations between the Databricks platform and your cloud account, and then Databricks deploys ephemeral compute clusters using cloud resources in your account to process and store data in object storage and other integrated services you control.

Our customers use Databricks to process, store, clean, share, analyze, model, and monetize their datasets with solutions from BI to machine learning. You can use the Databricks platform to build many different applications spanning data personas.

The Databricks workspace provides user interfaces for many core data tasks, including tools for the following:

- Interactive notebooks
- Workflows scheduler and manager
- SQL editor and dashboards
- Data ingestion and governance
- Data discovery, annotation, and exploration
- Compute management
- Machine learning (ML) experiment tracking
- ML model serving
- A feature store
- Source control with Git

In addition to the workspace UI, you can interact with Databricks programmatically with the following tools:

- REST API
- CLI

Databricks combines the power of Apache Spark with Delta Lake and custom tools to provide an unrivaled ETL (extract, transform, load) experience. You can use SQL, Python, and Scala to compose ETL logic and then orchestrate scheduled job deployment with just a few clicks. Databricks provides a number of custom tools for [data ingestion](#), including [Auto Loader](#), an efficient and scalable tool for incrementally and idempotently loading data from cloud object storage and data lakes into the data lakehouse.

BigQuery:

BigQuery is a fully managed enterprise data warehouse that helps you manage and analyze your data with built-in features like machine learning, geospatial analysis, and business intelligence. BigQuery's serverless architecture lets you use SQL queries to answer your organization's biggest questions with zero infrastructure management. BigQuery's scalable, distributed analysis engine lets you query terabytes in seconds and petabytes in minutes. BigQuery interfaces include Google Cloud console interface and the BigQuery command-line tool. Developers and data scientists can use client libraries with familiar programming including Python, Java, JavaScript, and Go, as well as BigQuery's REST API and RPC API to transform and manage data. ODBC and JDBC drivers provide interaction with existing applications including third-party tools and utilities.